Solar Powered Smart Mini Cold Storage System for Fresh Vegetables in North Eastern Region (NER)

Submitted By:

Pratik Kumar Jha

Vinay

Rakshit Goutam

Kartikya Mandal

Piyush Kumar Rai

Khushi Jain

Abstract:

This project presents a **Solar-Powered Smart Mini Cold Storage System** designed to provide an affordable and portable solution for preserving fresh agricultural produce. The system utilizes the **Peltier effect**, a thermoelectric phenomenon in which a temperature difference is produced when electric current passes through a semiconductor device. Peltier modules are integrated with an insulated storage chamber, heatsinks, and cooling fans to achieve effective heat transfer and cooling. **Solar energy** is used as the primary power source, while a rechargeable battery provides backup power during periods of low sunlight. An **ESP32-based monitoring system** with temperature and humidity sensors enables real-time monitoring of the storage environment. The proposed system is particularly aimed at addressing post-harvest losses in the **North Eastern Region (NER)**, where access to reliable and affordable cold-storage infrastructure can be limited due to difficult terrain, power availability, and transportation challenges. The project focuses on developing a **compact, decentralized, and scalable cold-storage solution** that can be deployed at farms, collection centers, cooperatives, and local markets. Through this project, we aim to combine renewable energy, thermoelectric cooling, and smart monitoring to improve the storage life of fresh produce and reduce post-harvest losses.

Introduction:

Refrigeration is essential for preserving fresh agricultural produce by maintaining a temperature lower than the surroundings. Unlike conventional refrigerators that use a compressor, condenser, evaporator, and expansion valve, our system uses **Peltier modules, heatsinks, and cooling fans** for heat transfer.

A Peltier module creates a temperature difference when electric current passes through it. Its cold side absorbs heat from the insulated chamber, while the hot side releases heat through a heatsink and fan. The system is further supported by **solar power, battery backup, and ESP32-based temperature and humidity monitoring**, making it a compact and suitable solution for decentralized cold storage.

Problem Statement:

The North Eastern Region (NER) produces a substantial quantity of fresh vegetables and horticultural crops. However, due to inadequate cold storage infrastructure, unreliable electricity supply, difficult terrain, and transportation delays, farmers often face heavy post-harvest losses. Most remote farming areas lack access to affordable small-scale cold storage facilities near production clusters and local markets. As a result, fresh vegetables deteriorate rapidly before reaching consumers, reducing farmer income and affecting supply chain efficiency. There is a need for a low-cost, energy-efficient, and decentralized cold storage solution suitable for rural and remote regions of NER.

Methods/Design Strategy:

- At first, we built a 3D-solidworks model of specified dimensions of the refrigerator unit. The volume of the inner compartment was 10400 cc. The dimensions were chosen minimally as to create enough cooling effect with a single Peltier module and to facilitate ease in placement.
- One of the main objectives of our project was to make the project lightweight and also cost-efficient. To satisfy this interest, we used cork sheet to build the main frame of the refrigerator.
- Cork sheets have air gaps within them. Thus, it also performs good as a heat insulator.
- The Peltier module was attached with two heatsinks on both of its surfaces. Two cooling fans were also attached along with the fans. As the heated surface was reducing the temperature of the cold surface, a large sized heatsink and a fan with higher rpm was used on the heated surface.
- The Peltier module along with heatsink and fans were installed on the top surface of the refrigeration unit. Thus, a mixed convection effect was created inside the refrigeration chamber. A schematic diagram of the whole cooler module is given below.

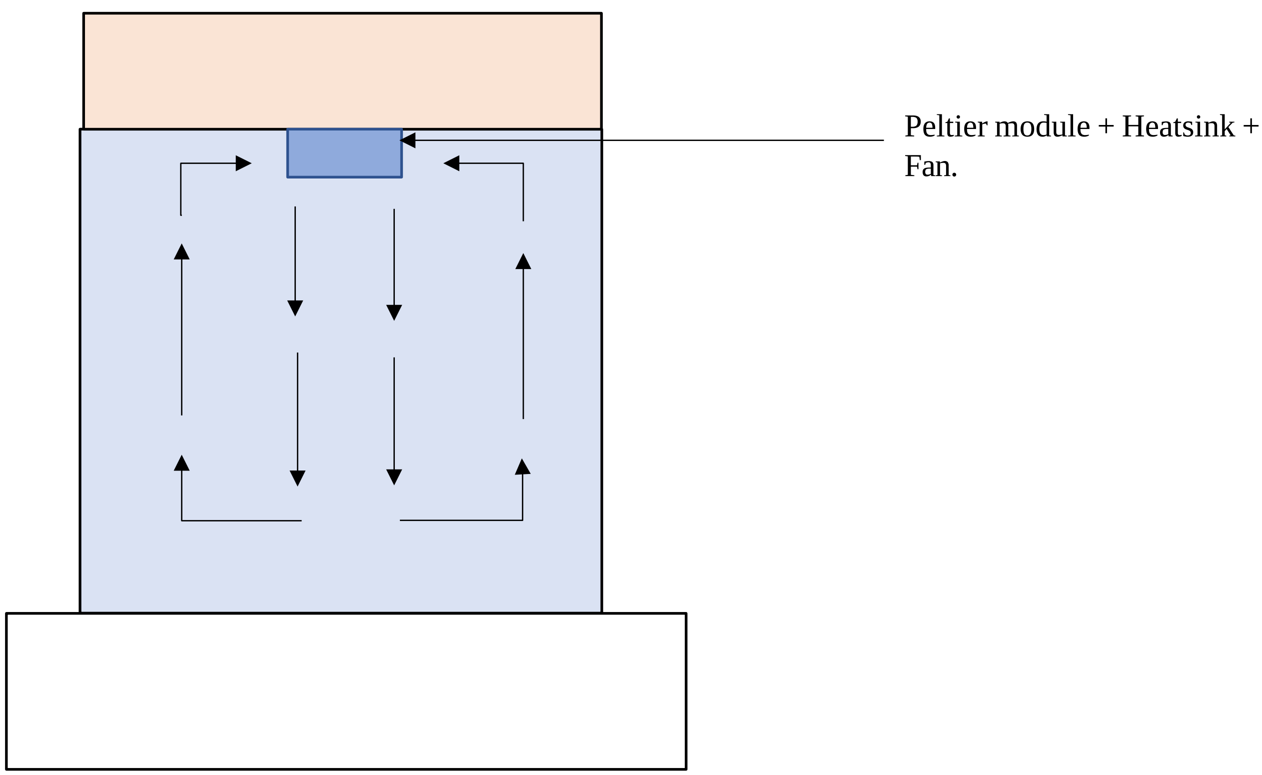


Figure 1: Schematic Diagram of cooler module.

Components:

Peltier Module:

Peltier module TECL 12706 is used in our project. The module can be seen in figure 2:
Specifications of the module is as follows:
Operating Voltage: 12 V
Maximum Voltage: 15.4 V
Maximum Current: 6A
Maximum Power: 92 W
Maximum Temperature: 138^{o} Celsius
Surface area: 40 * 40 mm.

Figure 2: Peltier Module

Heat Sink:

A heat sink is a component that increases the heat flow away from a hot device. There are two types of heat sink passive and active heat sink. We have used a 40*40 mm Aluminum heat sink and a round heat sink of 90mm diameter. Heatsink used in this project can be seen in figure 3.

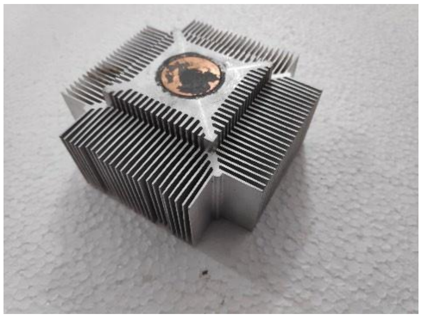


Figure 3: Heat Sink

Cooling Fan:

CPU cooling fan is any fan inside or attached to a computer case used for active cooling. Both axial and centrifugal fans are used in our project. We have used two DC-12 V fans in our project. The speed of our fan is 1200 rpm and 2000 rpm. The cooling fan used in this project can be seen in figure 4.

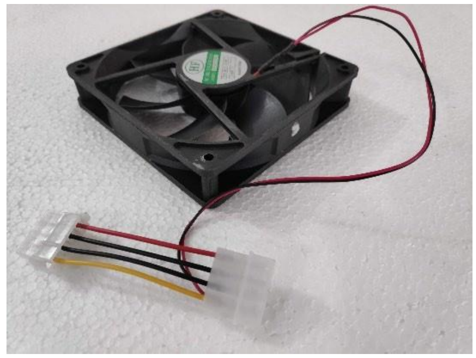


Figure 4: CPU Fan

L7805 voltage regulator:

Voltage regulators are used in power supply for stabilizing the DC voltages. L7805 is a three terminal voltage regulator IC. The voltage regulator used in this project can be seen in figure 5.

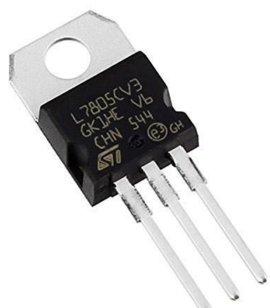


Figure 5: L7805 Voltage Regulator

Thermal Paste:

Thermal paste or some oily thermal interface is necessary because it fills in the microscopic imperfections that otherwise trap air particles between the Peltier and heat sink preventing the Peltier from properly cooling. Figure 6 shows the thermal paste we used in this project.

Figure 6: Thermal Paste

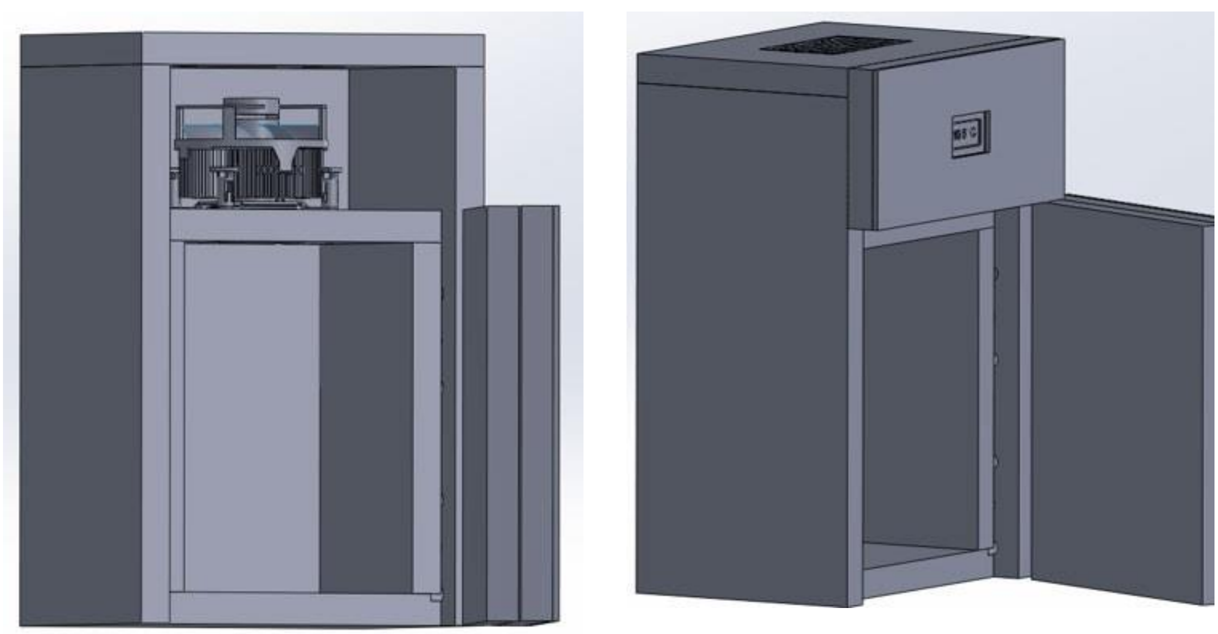
Cork sheet:

One of the main advantages of using cork to build the body of a product is its low cost. Cork is a renewable resource that is relatively easy to harvest and process. As a result, cork products are typically very affordable. That's why we used cork sheet to build the body which can be seen in figure 7.

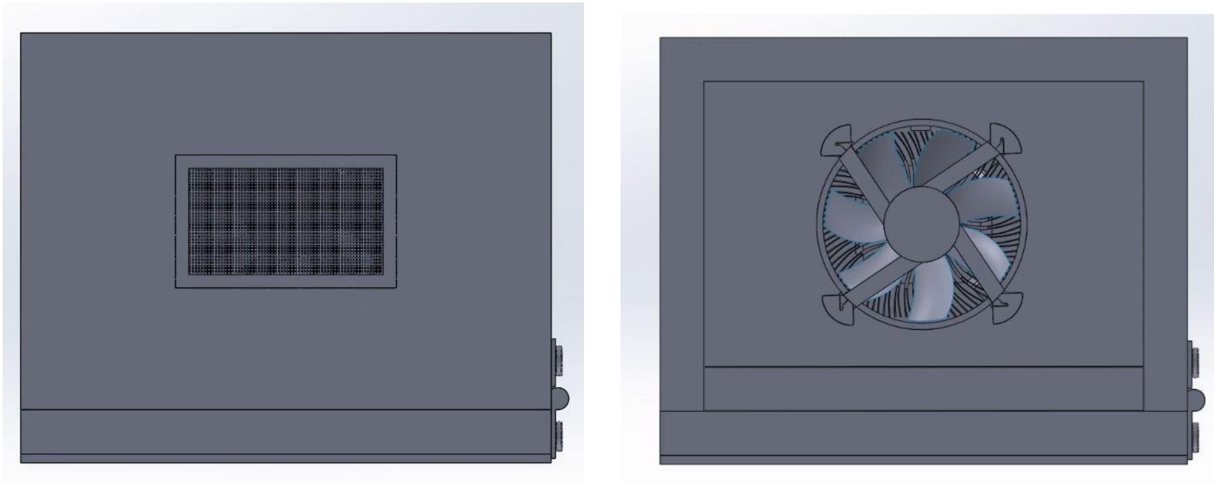
Figure 7: Cork sheet

SolidWorks Model:

SolidWorks 2019 software is used to 3D model our current problem. The isometric view and top view of the model can be seen in figure 8.



Isometric Views



Top Views

Figure 8: SolidWorks model of current project

Actual Model:

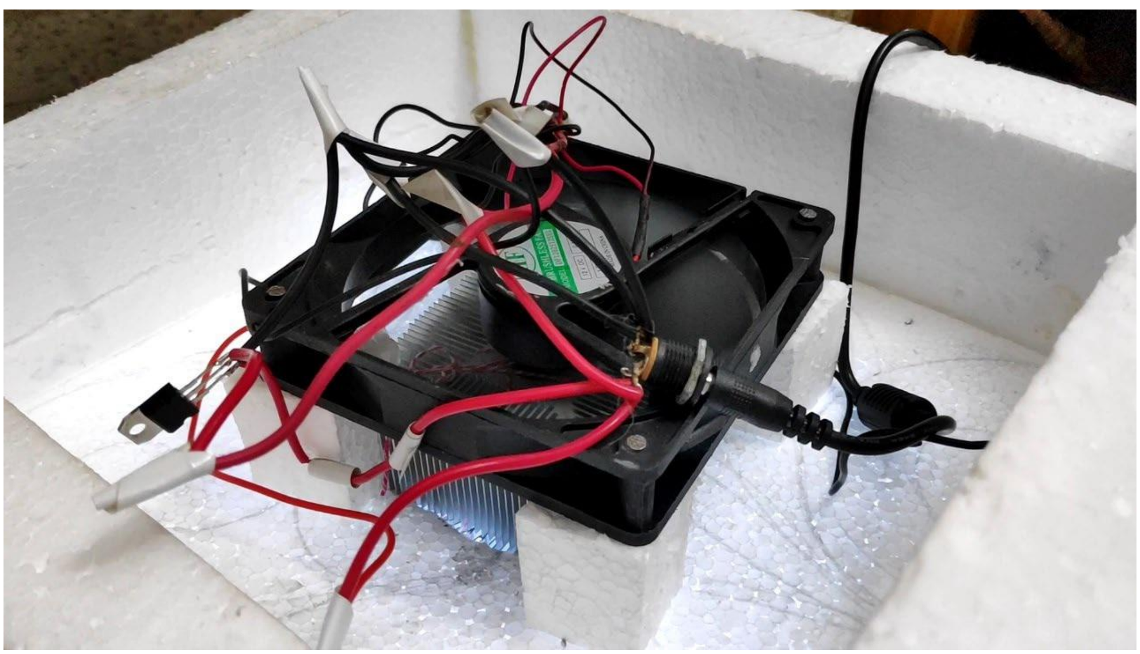


Figure 9: The Cooler Module

Figure 10: Front View of the Refrigerator

Figure 11: Inside of the refrigerator

The lowest temperature we were able to achieve is $15.2^{o}C$ which can be seen in figure 12.

Figure 12: Lowest temperature recorded.

Calculation:

Coefficient of Performance:

Generally, COP is determined by the ratio between energy usage of the compressor and the amount of useful cooling at the evaporator.

$$COP = \frac{\text{Desired Effect}}{\text{Work Output}} = \frac{Q_1}{W}$$

Maximum COP by Carnot refrigerator,

$$COP = \frac{T_{cold}}{T_{hot} - T_{cold}}$$

For our refrigerator,

$$COP = \frac{(0.0126)(1.005)(1000)10}{40*5*6} = 0.0106$$

Justification:

The performance of our refrigerator is suboptimal due to several factors. First, we used a standard CPU cooler instead of a specialized cooler designed for 7400 cc of air. Second, we were limited to using a single Peltier module due to power constraints. Third, we were also constrained by financial limitations. To improve the performance of our refrigerator, we would need to use specialized components. This would allow us to scale the system and achieve better results.

Problem faced during the project:

- Aluminum foil is a poor insulator and so it keeps food warmer for long and also keeps things cold. It acts as a barrier to air and oxygen. So, we have used aluminum foil to reduce the heat loss which can be seen in figure 13 due to structural defects. But aluminum foil also reflects radiation what we didn't notice earlier. After attaching aluminum foil, we noticed that the temperature in our refrigerator increased instead of decreasing. So, we had to remove it completely and redesign the interior.
- We had to be more careful and conscious before introducing innovation due to limited time and resources in this COVID outbreak.
- We used digital thermometer to measure temperature, but it is not dedicated for measuring the gas temperature. That's why we did n't get the exact temperature drop.

Figure 13: Using aluminum foil hampered the overall performance.

Recommendation:

- Instead of using 12V AC/DC adapter, 12V Li-Po rechargeable battery could have been used for portability.
- The cork sheet frame was quite delicate. Wrapping the outside surfaces with hard board would provide some rigidity to the frame.
- The inside chamber was not properly insulated. Proper insulation techniques may provide a better performance.
- Multiple Peltier module can be installed to increase cooling effect even though cost would increase subsequently.

Conclusion:

Collaborating on a heat transfer equipment project was a valuable learning experience. We also gained an understanding of thermoelectric devices. Completing the project during the pandemic was a significant accomplishment. We will improve the performance of our mini refrigerator in the future by addressing its shortcomings.